

NAVCORE-SoOP

Order to first flight

The complete runbook, with an audit of what is still manual,
soldered by hand, or not yet measured

Board	45.1 × 47.3 mm, 6-layer, STM32H743VIT6, Revision C
Gate verdict	READY TO ORDER — every gated prerequisite proven (the gate prints the count)
Hardware verdict	NOT READY TO FLY — 6 items need hardware, none of which can change what is ordered
Toolchain	KiCad 9.0.8, ArduPilot Copter-4.7.0, Python 3 + pcbnew

*Every figure in this document was re-derived from the repository's own tooling on 19 September 2026 rather than copied from another document. Where a number is a judgement, an assumption or genuinely unknown, it says so. Sections marked **flag** are the ones capable of costing money, a board, or a bench session.*

Contents

1	How to use this, and how to verify it	3
1.1	The four commands that matter	3
1.2	Freshness	3
2	Part 1 — Order the bare board and its assembly	3
2.1	Regenerate, do not trust	3
2.2	What to upload	3
2.3	Fabrication settings	4
2.4	Quantity, and the tier decision	4
2.5	The variant files, and why two of them are identical	4
2.6	Parts worth checking before you pay	4
2.7	The DNP list — derived, never typed	5
2.8	Check these on screen at upload	5
2.9	Paying	5
3	Part 2 — Order the parts: the board's bench first, then Iridium, then the aircraft	5
3.1	The one order — nothing here should make you order twice	7
3.2	The bench measurements that could have cost you a purchase	8
4	Part 3 — While you wait (free, and the highest-value work)	9
5	Part 4 — On arrival: inspect before you spend more	9
5.1	Visual inspection	10
5.2	Confirm the machine-readable facts	10
5.3	The frame measurements that settle a real decision	10
5.4	The three things no drawing can answer	10
5.5	T0 — the mechanical checklist, before any loom is cut	10
6	Part 5 — Mechanical assembly	11
6.1	Stack height	11
6.2	The two screw warnings — read both	11
6.3	Screw heads against copper — 0.08 mm	11
6.4	Sequence	11
6.5	What plugs in where	12
6.6	The pyro channel (lander only) — arming is not optional	13
6.7	The harness — six runs, and how not to make a nest of them	13
6.8	Things that will bite you	13
6.9	J2, the ESC connector — confirmed rather than assumed	14
7	Part 6 — Soldering: four joints minimum to fly	14
8	Part 7 — T1: first power, before the MCU sees anything	14
9	Part 8 — T2: MCU alive	15
10	Part 9 — T3: sensors on the bench, props OFF	15
11	Part 10 — T3a: thermal, and the board measures half of it for you	16
12	Part 11 — T3b: the RF chain, and the single open question	17
13	Part 12 — T3c: IMU temperature calibration	17
14	Part 13 — T3d: the seven unlabelled pads	18

15 Part 14 — T4/T5: tethered, then flight, strictly in this order	18
16 Part 15 — Audit: what is still manual, soldered, or unmeasured	18
16.1 15.1 Manual soldering	19
16.2 15.2 Manual assembly	19
16.3 15.3 Measurements that have not been confirmed	20
16.4 15.4 The three worst, ranked	21
16.5 15.5 Design-level inefficiencies, for a future revision	21
17 Part 16 — What only the outside world can settle	22
18 Part 17 — What is not built yet	22
19 The one-page order	23

1 How to use this, and how to verify it

This is one document in the order you actually do things: order the board, order the aircraft parts, assemble, power it up, and fly it. It exists because the same information used to be spread across four draft documents (ORDER, ASSEMBLY, BUILD and BUYING) that had drifted apart — in three cases far enough to send you shopping for parts with nowhere to put them. On 2026-09-19 the first three were folded into this one and deleted; docs/BUYING.md survives as the order sheet generated from docs/PARTS.csv, the decision record.

Verify before you spend. Read the verdict, not a summary:

python3 tools/preflight.py → must print **READY TO ORDER**

The verdict is *derived* from a manifest of gated prerequisites, so an unrun tool, a non-zero exit, a failed check or an unclassified manifest entry is a blocking failure. It cannot be typed by hand and it cannot shrink silently. What it proves is that the *design* is orderable. It cannot prove anything about money, stock, or JLCPCB's own upload preview — those are Sections 2.8 onward.

1.1 The four commands that matter

python3 tools/preflight.py	the gate; must say READY TO ORDER
bash tools/make_order_bundle.sh	regenerate fab/ and zip the order
bash tools/build_firmware.sh	build arducopter.apj, pinned to Copter-4.7.0

1.2 Freshness

fab/stock-snapshot.json is re-fetched from jlcpcb.com and tools/check_stock.py blocks the gate once it is older than **7 days**. It was last refreshed 2026-09-19 and resolves **56 of 56** non-DNP codes with cover for two boards. Everything else in fab/ is regenerated by tools/make_order_bundle.sh, which refuses to build the bundle if the copper layer count is not 6.

2 Part 1 — Order the bare board and its assembly

2.1 Regenerate, do not trust

bash tools/make_order_bundle.sh

This re-exports the gerbers and drill, re-runs gen_bom.py for *all three* variant pairs, checks the layer count, and writes fab/NAVCORE-SoOP-order-bundle.zip.

Do not hand-copy or hand-edit anything in fab/. A cp of a stale variant into the bundle is exactly how the Economic BOM once shipped quoting Y2's superseded, unbuyable part code for three days — and the order notes of the day were recommending those files at the checkout while it did.

2.2 What to upload

File	Where
fab/gerbers/ — zip the <i>contents</i>	PCB gerbers + Excellon drill
fab/BOM-NAVCORE-SoOP.csv	assembly BOM, the default build
fab/CPL-NAVCORE-SoOP.csv	pick-and-place, the default build

Or upload the bundle zip, which contains those plus HOW-TO-ORDER.txt. fab/gerbers/ holds **16 files**: 14 .gbr (6 copper, 2 paste, 2 silkscreen, 2 mask, edge cuts, drill map), NAVCORE-SoOP.dr1, and NAVCORE-SoOP-job.gbrjob.

Stop-condition at upload. If the preview shows **four** copper layers, abort. The power planes are missing and the board will not work. It must show six.

2.3 Fabrication settings

Setting	Value	Why
Layers	6	In1 is a solid GND plane; In4 carries the rails
Dimensions	45.1 × 47.3 mm	KiCad’s box; JLCPCB shows 45.0 × 47.2
Thickness	1.6 mm	30.5 mm stack standard
Outer copper	1 oz	every current figure assumes it
Surface finish	ENIG	0402s and LGA-14s want a flat finish
Min track / spacing	0.1016 mm (4 mil)	should be free tier
Min via / drill	0.45 mm / 0.20 mm	
Impedance control	No	USB is a plain pair; the RF run loses <1.5 dB
Via covering	Epoxy Filled & Capped	31 vias sit in SMD pads
Stencil	not needed	everything machine-placed but Section 7
Mark on PCB	Remove Mark	no JLC marker on the board; small fee

The 1 oz line is not cosmetic: **every current-capacity figure in `check_build.py` assumes it**, and 0.5 oz halves each trace’s rating.

2.4 Quantity, and the tier decision

Order 5 bare boards and 2 assembled. Five is the multilayer minimum; two is the usual SMT assembly minimum.

	Standard (panelised)	Economic
Board minimum	70 × 70 mm , so 2 × 2 panel + rails	10 × 10 mm, no rails
Places	everything, including U3	everything except U3
Feeder fees	included	GBP 40–46 still charged
Hand-fitting	none	U3 (ICM-42605, second IMU)

Recommendation: Standard, panelised 2 × 2, “panel by JLCPCB”. Do not commit a panelised board file — the board files stay exactly as gated, so nothing has to go through DRC, connectivity and the whole gate again. A hand-panelised `.kicad_pcb` would have to, for no benefit.

If you take Economic instead, **take its BOM and CPL as a pair** (`*-economic.csv` for both). Mixing an Economic BOM with the default CPL hands JLC a placement list for parts the BOM does not order. `tools/check_variants.py` — now a gated check — fails if any pair disagrees with the board.

2.5 The variant files, and why two of them are identical

Variant	BOM lines	Placements	Difference from default
NAVCORE-SoOP (default)	70	162	—
NAVCORE-SoOP-economic	70	161	omits U3 only
NAVCORE-SoOP-nofpv	70	162	none — byte-identical

`-nofpv` is byte-identical to the default pair: Revision C removed the 9 V VTX block it used to omit (L5 is now the payload buck’s inductor, and a VTX runs from `+5V_PAYLOAD`). It is a label, not a saving. Choosing Economic only makes sense if you want to hand-fit the second IMU.

2.6 Parts worth checking before you pay

Stock is no longer quoted in prose. It lives in the dated snapshot `fab/stock-snapshot.json`, age-gated at 7 days by `tools/check_stock.py`.

Two substitutions are **not optional**:

- **Y1 must be C2682774** — a *passive crystal*, CL 20 pF. The visually identical C2901556 is an **active oscillator**: same SMD3225-4P land pattern, but pin 4 is VDD, which this board grounds. Substituting it gives a board that never starts.
- **Y2 must be C5563878**. Its predecessor C22381771 was down to **1 unit** with no published restock date when `check_stock.py` was written.

2.7 The DNP list — derived, never typed

One designator ships unpopulated: R15, the 120 Ω CAN bus terminator. A CAN bus is terminated at its two *ends*; fit R15 (0402, across CANH and CANL) only if this board is the last node on the bus. Revision C fits everything else, including the CAN transceiver U11 (on its own +3V3_CAN LDO, U21, fed from the payload rail) and the payload buck U20.

Do not type this list — `make_order_bundle.sh` derives it from the BOM into `HOW-TO-ORDER.txt`.

2.8 Check these on screen at upload

- Outline and board size: **45.1 × 47.3 mm**
- Copper layer count: **6**
- Drill count and the 0.20 mm minimum
- **The placement preview: nothing should need turning.** Every rotation and centre in the CPL is fitted to JLCPCB's own footprint for that LCSC part (`tools/jlc_orientation.py`, 160 of 163 agreeing with the rule-based rotation it replaced). A part that still looks turned is a finding — report it rather than rotating it by hand
- **1 oz outer copper**, explicitly
- **Panel by JLCPCB.** Standard assembly requires a single PCB of at least 70 × 70 mm and this board is 45.1 × 47.3 mm, so the panel and its breakaway rails have to come from somewhere. Their own panelisation is the low-risk answer: you then confirm the panel outline in the same preview as the CPL.

2.9 Paying

- **Coupons are once-per-order** and cannot be combined. Take the SMT one.
- **6-layer promotions recur**; worth waiting a few days if you are not in a hurry.
- **UK import VAT is 20%**, collected by the courier above GBP 135 with an €8–12 handling fee. It is not a JLCPCB charge and is easy to forget when comparing against a UK-stocked alternative.

3 Part 2 — Order the parts: the board's bench first, then Iridium, then the aircraft

The list is staged by what you are trying to prove, not by what the aircraft needs. **A1 and A2 need no aircraft at all**: they power, flash and bench the board, and put an Iridium receive chain on the bench and then on the board. **B1 and B2 are the aircraft**. Order everything at once if you can — lead times run 2–4 weeks — but if the budget is staged, this is the order. The table is generated from `docs/PARTS.csv` (`python3 tools/gen_doc_tables.py`); `docs/BUYING.md` is the same list with provenance and price tags.

A1 — The board, and its bench

7 lines, about GBP 257. Power it, flash it, prove it - no aircraft. Runbook Parts 7-10 (T1-T3a).

Item	Qty	GBP	Source
NAVCORE-SoOP PCB + assembly	1 order	190	JLPCB
ST-Link V2 programmer	1	4	AliExpress
microSD card, 32GB	1	8	AliExpress / anywhere
USB-C cable, data-capable	1	3	anywhere
JST-SH 1.0mm 8-pin pigtail (spare J2 ESC lead)	2	3	AliExpress
Bench power supply, adjustable with CURRENT LIMIT	1	35	AliExpress
Thermocouple meter or IR thermometer	1	14	AliExpress

Already have: Soldering iron and basic tools, Multimeter with continuity buzzer.

A2 — Iridium navigation, no aircraft

6 lines, about GBP 96. The receive chain the board cannot supply, and the bench reference receiver for T3b step 1. The tuner (U13) is on the board; this is what feeds it.

Item	Qty	GBP	Source
Nooelec SAWbird+ IR (1620MHz LNA+SAW)	1	35	Nooelec
Nooelec 1620MHz Iridium patch antenna	1	25	Nooelec
SMA-male to U.FL (IPEX MHF1) coax pigtail, 10-15 cm	2	4	AliExpress
micro-USB power lead for the SAWbird+ (5 V from PV1/PV2)	1	2	AliExpress
RTL-SDR Blog V4 (R820T2, with bias tee)	1	30	AliExpress / RTL-SDR Blog
gr-iridium + iridium-toolkit (software)	1	0	GitHub (mucce)

Already have: Raspberry Pi 5.

B1 — The aircraft - minimum to fly on GPS

22 lines, about GBP 333. Frame, propulsion, radio, GPS, fasteners, looms. Every optional sensor ships disabled, so nothing in B2 is needed for a GPS flight.

Item	Qty	GBP	Source
TBS Source One V5 7in DC	1	36	HobbyRC UK
Motors 2806.5 1300KV	4	60	AliExpress
Props 7040 or 7042	4 sets	14	AliExpress
SpeedyBee BLS 60A 4-in-1 ESC	1	48	AliExpress
RadioMaster Pocket ELRS transmitter	1	68	AliExpress
ELRS 2.4GHz receiver	1	15	AliExpress
M10 GNSS + QMC5883L compass (antenna INTEGRATED)	1	24	AliExpress
5V PASSIVE piezo buzzer	1	3	AliExpress
WS2812B LED strip	1	6	AliExpress
XT60 pigtails + silicone wire + heatshrink	1 kit	12	AliExpress

Item	Qty	GBP	Source
Smoke stopper (XT60)	1	10	AliExpress
EC5 to XT60 adapter	2	6	AliExpress
Landing gear - TPU skids (3D PRINT)	1 set	0	3D print
Landing gear - TPU/nylon skids, BOUGHT (backup for the printed set)	1 set	8	AliExpress
M3x14 socket screws (motor)	16	3	AliExpress
M3 female-female standoffs 12 mm (ESC-to-FC spacer)	4	2	AliExpress
M3x30 socket screws (stack bolts)	4	2	AliExpress
M3 socket screw assortment (10/12/14/16/20 mm) + M3 washers	1 kit	5	AliExpress
M3 silicone grommets, spare set (30.5 mm stack)	1 pack	2	AliExpress
JST-SH 1.0mm 4-pin pigtails (the sensor and RC leads)	1 pack of 10	5	AliExpress
JST-GH 1.25mm 6-pin spare lead (J3 GPS port)	1	2	AliExpress
JST-SH 1.0mm 3-pin pigtail (J15 LED strip)	1	2	AliExpress

Already have: 4S LiPo packs, LiPo balance charger.

B2 — Aircraft sensors and payload - optional

5 lines, about GBP 68. RNGFNDn_TYPE and PRX1_TYPE ship at 0; enable each one at the bench after it is fitted. The +5 V rail cannot carry all of these AND the SAWbird+ at once - check_modules.py prints the budget.

Item	Qty	GBP	Source
Benewake TFS20-L lidar	1	32	AliExpress (or 3DXR UK GBP35.70)
VL53L1X ToF module (GY-53-L1X)	1	6	AliExpress
LDROBOT LD06 360 lidar (INDOOR USE ONLY)	1	14	eBay - Okdo Lidar Hat kit (use the LI- DAR UNIT only)
XIAO ESP32S3 Sense (recording camera)	1	14	AliExpress
Camera power: thin silicone wire + 2-pin JST pigtails	1 kit	2	AliExpress

Total, everything to buy: about GBP 754. Generated from docs/PARTS.csv by tools/gen_doc_tables.py; every row's reasoning and link is in the CSV, and docs/BUYING.md is the same list with prices and provenance tags.

Do not buy a bare PMW3901 or VL53L1X chip to hand-fit. Both were deleted from the board in the Rev B re-layout and have no pads; docs/BUYING.md priced a “GBP 227 hand-fit route” around them until 2026-09-19. The off-board GY-53-L1X *module* for the upward rangefinder is a separate, later purchase and is unaffected.

3.1 The one order — nothing here should make you order twice

Every “measure this, then use a different part” branch in this document is closed by buying the alternatives up front. The premium is about **GBP 6**.

What to add	What it closes
M3 socket assortment + washers	Both derived joints round <i>up</i> past their engagement target (4.5 mm and 6.7 mm into the thread) and the skid's tapped depth is the only measurement that can invalidate the length you picked.
4 × 12 mm female-female spacers	the ESC-to-FC separation is 12.1 mm and <code>fasteners.py</code> says plainly that a grommet cannot hold it. Without these the stack cannot be assembled at all.
the kit's own standoffs	do <i>not</i> buy 35 mm standoffs for the top plate; it has no 30.5 mm holes.
spare JST-GH 6-pin lead	the residual J3 risk is a <i>bulky moulded boot</i> on the lead you receive, not the board: the 5.30 mm clear run meets its derived 3.6 mm requirement. Test-fit the plug before soldering; a second lead is then the fallback rather than an interruption.
10 JST-SH pigtails, not 4 SAWbird+ , the Blog V4 dongle , the coax pigtail and the power lead , all with the first order	the first one you crimp badly is then not a project stall. T3b cannot be started without the first two, and the SoOP chain cannot be <i>flown</i> without the last two — this document listed neither until 2026-09-19. <code>docs/PARTS.csv</code> now carries all four

Three things need the parts in hand: the frame kit's top-plate standoff spacing (the kit supplies it), the camera mount (deferred, and the XIAO is the camera it must take), and the Radxa Zero 3W's hole pattern (deferred *with the companion* — the vendor never published it, so the tray is a print against a measured board). Note the asymmetry: the Radxa and the flow camera are deferred *purchases*, deferred because J4 does not exist to connect them, while the XIAO records to its own microSD over two wires and so is ordered now.

3.2 The bench measurements that could have cost you a purchase

Sort the bench items by what a result could move, because only one kind can reach your wallet. `python3 tools/check_bench_bounds.py` does this sort now and the gate walks its verdict, so the answer is printed rather than argued. Nine items became six on 2026-09-20, in five kinds:

- **CLOSED** — the answer was already fixed by a *document* or by the artwork, so there is nothing left to measure that could change the board. Three items left the list this way: the grommet, `BATT_AMP_PERVLT` and the J2 pin order. They are not deleted — `check_bench_bounds.py` still prints each with its evidence, and a reconciliation step refuses if this list and the manifest ever disagree.
- **BOUND** — the design-side corner is already under the limit and the measurement *confirms* it. U8: 87 °C against 125 °C. U9: 104 °C on good copper against 150 °C, on a board `thermal_vias.py` measures at 7443 mm² of plane. The grommet: the *metal* M3 head clears copper at every hole by +0.08 mm, and the flange is always the wider of the two, so a wider grommet adds only silicone bearing on mask.
- **PARAM** — the result is a *firmware constant*: `FLOW_ORIENT_YAW`. The board carries the pin and has no value to change.
- **MODULE** — the property belongs to a *bought part* that plugs in: the RF front end, which is the Nooelec SAWbird+ IR, and flow quality, which is the off-board camera. A shortfall is placement, shielding or a different module.
- **DATA** — the T3b record asserts nothing about the board at all.

The consequence is the one that matters before you pay: **no bench result can force a different board**. Every item is still measured, but it confirms a bound instead of discovering one, so the order does not wait for the session. The old text here warned that a hot U8 meant a respin; that risk was

closed on 2026-09-20 by the part swap (Part 10), which moved the worst corner to 87 °C and left it on the board.

Three are now closed outright rather than bought around. Each names the evidence:

Item	What fixed the answer, so nothing is left to measure
the grommet flange	the flange and the M3 cap head are <i>concentric</i> , so the flange is always the wider of the two and the metal always the narrower. A wider grommet adds only silicone bearing on mask; it cannot make the metal reach copper. Z was already swept 2.0–4.5 mm by <code>check_fit.py</code> , which requires every purchase to cover the range, so the design is <i>invariant</i> over the grommet
BATT_AMP_PERVLT	the ESC’s own manual publishes Scale=400, i.e. 40 mV/A, i.e. 25.0 A/V — and <code>defaults.parm</code> ships exactly that. The value was typed in two places, so <code>gen_hwdef.py</code> now <i>derives</i> it from <code>design.ESC</code> and <code>check_build.py</code> asserts it against the shipped file. Mutation-tested with the retired 52.7
the J2 pin order	<code>design.ESC</code> now <i>declares</i> the manual’s order and <code>check_build.py</code> holds the netlist against it. Nothing checked this before: J2 is a passive connector, so DRC, ERC and the pin-semantics check all pass a transposed one. The symptom is four motors on the wrong signals

The honest residue: no bench item can change what you *order*, and that is now a gated claim rather than a reading of this section — `check_bench_bounds.py` fails if any item is unclassified, if a bench entry is added without one, or if this list and the manifest disagree. *One* item has no analytic bound and stays a flight gate: whether the RF link closes depends on real sky and real desense, which no desk calculation settles. The one risk that *was* real has been closed: the old U8 corner of 139 °C was a budget error, and on 2026-09-20 (Part 10) the part was swapped in place to the LMR33630ARNXR, taking the worst corner to 87 °C against 125 °C. **Three spare bare PCBs** stay on the order for misadventure, not for a known unknown. `tools/fasteners.py` prints this list itself under “*not derived — these need the parts in hand*”.

4 Part 3 — While you wait (free, and the highest-value work)

1. **Register ArduPilot board ID 7180.** Not needed to order; needed before sharing. The request is written out in `firmware/BOARD-ID-REQUEST.md`, with the registry check, the alternates and the issue text. **Do not request 9001** — it is free but it is above the #7199 ceiling that `board_types.txt` explicitly asks you not to cross, while 4544 free IDs sit below it.
2. **Read the two screw warnings in Section 6.2 twice.** They are the ones that strip a frame.
3. **Print `docs/img/padmap.svg` and the assembly drawings `fab/assembly-top.svg` and `fab/assembly-bottom.svg`.** The silkscreen names every connector by function (ESC, GPS, RC, CAN, I2C, UART2, SERVO, PYRO, TOUCH, LED, BUZZ, SWD, USB, ANT, FLOW) and prints only the labels that sit unambiguously beside their part; every reference designator is on the assembly drawings. Seven pads arrive with no label (Section 14).
4. **Decide the USB edge orientation before cutting any loom.** All four edges carry a connector; there may be exactly one orientation that works.
5. **Dry-run the SITL suite** if you have a Linux box: `bash sitl/run_scenarios.sh`.

5 Part 4 — On arrival: inspect before you spend more

5.1 Visual inspection

Under magnification, look at U1 (**LQFP-100, 0.50 mm pitch**), U2/U3 (**LGA-14**, no accessible leads), and U13 (**TQFN-28 with an exposed pad**). You are looking for bridges and tombstones. JLC is good at this. Look anyway.

5.2 Confirm the machine-readable facts

```
python3 tools/preflight.py    never mind the board — confirm DRC, ERC, gerbers and
                              BOM/CPL are all current
python3 tools/check_variants.py  every variant pair matches the board
```

5.3 The frame measurements that settle a real decision

Check with calipers	Parsed value
top plate width	42.50 mm — the battery is 47 mm and already overhangs 2.25 mm each side; there is no room for an upward ToF beside it
bottom plate M3 positions	$x = \pm 14.60$ and ± 11.00 , no 30.5 mm pattern — so the FC cannot bolt under the bottom plate; that idea is closed
which plate is top	inferred from length (160.26 mm). The one genuinely open row

5.4 The three things no drawing can answer

- **microSD withdrawal clearance** — depends how *you* mount it
- **grommet compression under load** — a property of the rubber
- **the real TPU gear's shape, hole pattern and strength** — the assembly is modelled as four legs at the motor pattern on assumed dimensions

5.5 T0 — the mechanical checklist, before any loom is cut

Every figure here is recomputed by `python3 tools/check_mechanical.py`; if the board has moved since this was compiled, run it rather than trust the table.

- **Centre-plate opening** clears 45.1×47.3 mm plus the plug protrusions below. The FC plate is 48.50 mm across the binding axis, so the board has only **0.55 mm** on the tight side in the worst-case orientation — Revision C grew the board 1.2 mm on that axis, and this was 1.15 mm before. Check it with the real plate.
- **J1 USB-C — the tight one.** Its mouth is flush with the board edge, and the plug needs **5.8 mm clear beyond the top edge** at z 25.9–29.1 mm. A standoff on another plate, a plate lip or the battery strap across that edge makes the port unusable, and it is the only way to flash the board.
- **J2 ESC lead** needs 3.6 mm clear beyond the left edge, and its body sits **1.02 mm** from the nearest grommet — the tightest clearance on the board. Route the lead away from that corner, not over it.
- **J3 GPS** plug stays inside the outline, with 5.30 mm to R21 against a derived 3.6 mm. Test-fit the received plug; a bulky moulded boot is the one thing that can spoil it.
- **microSD** ejects on the **bottom** side, toward the bottom edge, and needs about **9.8 mm** of travel through a gap only 3.0 mm tall. Assume you will pull the board to change the card, and decide now whether that is acceptable.

- **Board rotation.** All four edges carry a connector, so there may be exactly one orientation that works. Decide it before cutting a loom. The firmware assumes the silkscreen arrow (it points at the USB edge) faces the nose; any other way round needs `AHRS_ORIENTATION` set to match.
- **Grommets** must hold 3.0 mm between the ESC’s parts and this board’s under compression: 12.2 mm PCB face to PCB face, which is what the 12 mm spacer sets. The compressed height of the rubber is the one *assumed* figure in the stack.
- **Nothing metal wider than a cap head at any hole.** The nearest copper to a hole is TP20 at 2.83 mm: an M3 cap head (2.75 mm radius) clears by 0.08 mm and a 7 mm washer overlaps by 0.67 mm. A 6 mm silicone grommet flange overlaps by 0.17 mm too, but it is an insulator — the gate lists it as a bench item to measure, not a failure.
- **Skids match the 19 × 19 mm motor pattern**, not the frame’s 16 × 16.

6 Part 5 — Mechanical assembly

6.1 Stack height

30.3 mm against the top plate’s underside at 32.5 mm — 2.2 mm spare. The top plate rides on the frame kit’s **22 mm front standoffs**, which stand on the mid plate (bottom plate 2.5 + arms 6.0 + mid plate 2.0 = 10.5 mm, + 22 = 32.5 mm). The 30.3 mm is `check_mechanical.py`’s figure and includes J3’s mated plug; `check_build.py`’s 19.8 mm is bare parts only and is not the one the plate must clear. Earlier drafts said “22.3 mm against 25 mm, 2.7 mm spare” — a Mark4-era number, retired. There is *less* room than that, not more.

6.2 The two screw warnings — read both

```
python3 tools/fasteners.py
```

- **Motor to arm, skid sandwiched.** Needs 13.5 mm, so order M3 × 14. But rounding 13.5 → 14 drives **5.5 mm into a thread designed for 4.0 mm**. A motor boss is often only about 4 mm deep, and a **bottomed-out screw feels tight while clamping nothing** — that is the failure that throws a motor in flight. Measure the tapped depth before fitting sixteen screws. M3 × 12 plus a washer is the usual fix.
- **FC to ESC to frame, the 30.5 mm stack bolt.** Needs **27.3 mm**, so M3 × 30. Rounding 27.3 → 30 puts **6.7 mm** into the press nut. If the tapped depth is shallower than that the screw bottoms out and clamps nothing.

This is the whole reason the buy list carries an M3 assortment rather than one length. `free_island` is not involved; the point is that both joints are decided by a measurement you have not taken yet, so own every plausible length.

6.3 Screw heads against copper — 0.08 mm

No copper is under any screw head, and the margin is thin. The closest is TP20 at **2.83 mm** against a **2.75 mm** cap-head radius — **0.08 mm**, and that assumes **no washers**. A washer, or a head even 0.1 mm over nominal diameter, puts metal on copper. If you add washers, re-measure before trusting this line.

6.4 Sequence

1. **Inspect the boards** (Section 4.1).
2. **Print the assembly drawings and the pad map.** Seven unlabelled pads — Section 14.
3. **Dry-fit the FC into the centre plate.** Confirm nothing blocks the bottom edge where the microSD ejects.

4. **Buzz the ESC cable** both ends against the J2 pinout below, before anything is powered.
5. **Sit a skid under each arm**, holes aligned. Motor on top, wires inboard. Thread all four screws by hand before tightening any; tighten diagonally.
6. **The stack, bottom to top: ESC, grommets, FC.** Fit the silicone grommets into the FC's four mounting holes; ESC into the frame on its 30.5 mm pattern; FC on top with the silkscreen arrow toward the nose. If USB-C has to face somewhere else, turn the board and set `AHRS_ORIENTATION` (Yaw90, Yaw180, Yaw270) to the arrow's direction.
7. **4 × M3 × 30 through the stack**, with the four 12 mm female-female spacers setting the ESC-to-FC separation.
8. **The top plate on the kit's own standoffs** — 22 mm at the front, over the stack, 30 mm at the rear. Do not substitute longer ones for the front: 2.2 mm is the clearance, and it is the kit's geometry that gives it.

6.5 What plugs in where

Module	Port	Pin order, 1 → 4
GPS + compass	J3, JST-GH 6P	the module's own lead, latched
ELRS receiver	J5, JST-SH 4P	5 V, TX, RC_IN, GND
I2C rangefinders	J9, JST-SH 4P	5 V, SCL, SDA, GND
360 lidar (later)	J11, JST-SH 4P	+5V_PAYLOAD, TX, RX, GND
SoOP antenna chain	J12, U.FL	coax from the SAWbird+; its power from PV1/PV2
recording camera, FPV	PV1/PV2	+5V_PAYLOAD and GND, soldered
LED strip	J15, JST-SH 3P	+5V_PAYLOAD, data, GND — or solder to PL1–PL3
buzzer	J16, JST-SH 2P	+5 V, drain — or solder to PZ1/PZ2
CAN / DroneCAN	J6, JST-SH 4P	+5V_PAYLOAD, CANH, CANL, GND
SWD (fallback flash)	J19, JST-SH 4P	+3V3, SWDIO, SWCLK, GND
TVC servos (lander)	J17, JST-SH 6P	PWM7, PWM8, PWM9, PWM10, +5V_PAYLOAD, GND
pyro / e-match (lander)	J18, JST-SH 2P	VBAT_FUSED, drain — arming switch in series
touchdown switch (lander)	J20, JST-SH 2P	TOUCHDOWN, GND
companion (deferred)	P71–P74 + P44	5 V, UART4 TX, UART4 RX, GND; PPS on P44

Serial leads cross over. The board's TX goes to the module's RX and the board's RX to the module's TX: on J5, pin 2 (TX) to the receiver's RX and pin 3 (RC_IN) to its TX. A Pixhawk GPS lead on J3 is already crossed. DroneCAN nodes ship with JST-GH leads; J6 is JST-SH in the same pin order, so make a GH-to-SH lead.

Hobby servos on J17. Servo plugs are 3-pin 2.54 mm females: signal (orange, yellow or white), + (red, middle), ground (brown or black). Make a lead from a 6-pin JST-SH pigtail and two 3-pin male headers: J17 pin 1 (PWM7) and pin 2 (PWM8) to the two signal pins, pin 6 to both grounds. For two 9 g micro servos, pin 5 can feed both red wires. Anything bigger goes on its own 5–6 V BEC, with the BEC's ground joined to pin 6 and pin 5 left unconnected. A JST-SH contact is rated 1 A, and +5V_PAYLOAD budgets 0.5 A for J17. `lander.parm` already drives SERV07/SERV08; check the travel direction with the props off.

Mount the compass as far from the ESC and the battery leads as the frame allows. There is no magnetometer on the FC by design: one 15 mm from a 60 A 4-in-1 is useless.

J17 is six pins, not eight. Servo power is *pin 5 only*; pin 6 is GND. `docs/LANDER-COMPATIBILITY.md` described an 8-pin header with +5 V on pins 5 and 6 until 2026-09-22 — a loom built to that table shorts the payload rail to ground.

The payload rail is budgeted per mission. U20 and its inductor L5 carry 1.6 A continuous. The drone's payload (VTX, camera, SAWbird+, LD06, LED, CAN) is 1.17 A; the lander's (two TVC

servos, LED, CAN) is 0.76 A. **Wire one mission's payload, not all of it:** every payload connector at once is 1.67 A, over the inductor and about 147 °C at U20's worst corner. `check_modules.py` fails any declared mission that does not fit.

6.6 The pyro channel (lander only) — arming is not optional

J18 fires an e-match from VBAT_FUSED (3 A fuse F1) through the low-side MOSFET Q5. Between the MCU and a firing there is exactly *one* GPIO (PC5): R55 (47 kΩ) holds the gate off through reset and under stock MatekH743 firmware, and the NAVCORE firmware ships PYRO_FIRE OUTPUT LOW — but a single firmware fault still fires it. **Put a physical arming switch or a removable shunt plug in series with the e-match in the J18 lead, open until the vehicle is on the pad.** It is on `docs/PARTS.csv` for that reason. Do not bench-test the channel with an e-match connected; use a resistor or an LED.

6.7 The harness — six runs, and how not to make a nest of them

Run	Wires	Note
J2 → ESC	8	comes with the ESC; stays inside the stack
J3 → GPS	6	the module's lead; the compass rides on it
J9 → rangefinders	4	one lead, split at a junction: down and up
J5 → ELRS	4	
J12 + P41/P46 → SAWbird+	coax + 2	the antenna chain and its power
PZ, PL → buzzer, LEDs	2 + 3	the LED strip is optional

Four routing rules, one of them specific to this aircraft:

1. **Keep every signal cable and the Iridium coax away from the four motor leads.** They carry kilowatts switched at tens of kHz. On a normal quad that is good practice; on this one it is the difference between hearing Iridium bursts and not (T3b).
2. Route along the arms and plate edges, zip-tied every 40–50 mm. Slack belongs in one service loop near the stack.
3. Leave the **USB edge** clear (Section 5.5).
4. The ELRS antenna and the Iridium patch get separation from everything else, and from each other.

6.8 Things that will bite you

- The pack is EC5; everything else is XT60. Use the adapter — do not re-terminate a hard-case pack on build day.
- The frame offers 16 × 16 and 19 × 19 motor patterns. The motors are 19 × 19.
- Measure both tapped depths before committing sixteen screws.
- Seven pads are unlabelled, and two are supply pads: P41 is +5 V and PV2 is payload ground.
- The ESC cable is a gamble until you buzz it.
- J17 is six pins: servo power is pin 5 only, pin 6 is GND.
- The board is **5S maximum** — the input TVS sets it (Part 1).
- Wire one mission's payload onto +5V_PAYLOAD, not every payload at once.
- Confirm 1 oz outer copper on the order; 0.5 oz halves every trace's rating.

6.9 J2, the ESC connector — confirmed rather than assumed

Wired **1 GND**, **2 VBAT**, **3 M1**, **4 M2**, **5 M3**, **6 M4**, **7 CUR**, **8 TEL** (pins 9/10 are the shell tabs). Betaflight’s board page for the SpeedyBee F405 V4 gives the same FC-side order, so it **matches** — no cable rework needed.

The vendor table reads oddly because it lists the two *ends* of the cable in opposite order, since they face each other. Reverse the ESC column and every pair lines up: GND↔GND, BAT↔VBAT, M1↔S1, ..., CUR↔CURRENT.

Buzz it anyway. One automated read of a wiki table is not the same as measuring the cable you received. An 8-pin JST-SH is re-pinnable with a fine pick — a mismatch costs a cable rework, not a respin.

Pin 8 is inert. The ESC end is N/A — stock BLHeli_S has no telemetry output at all. SERIAL5_PROTOCOL 16 and the UART8_RX wire are correct and cost nothing, but they will receive nothing until a telemetry-capable ESC (BLHeli_32, AM32) is fitted. That is why the board ships INS_HNTCH_MODE 1 (throttle) rather than 3 (RPM). The upgrade path needs no wiring: flash **BlueJay** or **AM32** onto the ESC over BLHeli passthrough (SERVO_BLH_AUTO 1 already enables it), set SERVO_BLH_BDMASK 15 for bidirectional DShot on motors 1–4, then INS_HNTCH_MODE 3 — the RPM-tracked notch is the single biggest flight-quality win available on this aircraft.

7 Part 6 — Soldering: four joints minimum to fly

Everything else plugs in.

- **Buzzer** — plug into J16, or solder to PZ1 (+) and PZ2 (−). Must be a **passive** piezo: PA15 is a timer channel and ArduPilot generates the tone patterns itself. An active buzzer has its own oscillator and loses every arming and failsafe pattern.
- **LED strip, optional** — plug into J15, or solder PL1 = data, PL2 = +5V_PAYLOAD, PL3 = GND. **PL1 and PL2 have no label.** 5 V WS2812B. One landing, not both.
- **P41 and PZ1 are +5 V and unlabelled; PL2 is +5V_PAYLOAD and unlabelled.** The camera’s and the SAWbird+’s power leads land on PV1/PV2 (the payload rail), *not* on P41. Identify every pad on the printed gerber before the iron touches it — never by counting from a neighbour.

8 Part 7 — T1: first power, before the MCU sees anything

USB disconnected. No battery. Do not skip this. A board that goes straight to USB can destroy the MCU on a rail fault that a current-limited supply would have caught for nothing.

Bench supply on VBAT, **16.8 V**, **current limit 100 mA**. VBAT_IN reaches only J2.2 — there are no separate battery pads — so feed it through the bare-ended 8-pin JST-SH pigtail from the order (J2.1 GND, J2.2 VBAT). Powering through the ESC instead works, but the ESC’s own draw then sits inside the 80 mA below.

#	Check	Expect	If wrong
1.1	current at power-on	< 80 mA	something is shorted: kill it and inspect U8, U20 and the bulk caps
1.2	+5V rail	5.02 V ±0.2	U8 LMR33630ARNXR, L2, divider R6/R7
1.3	+3V3 rail	3.3 V ±0.1	U9 AP2112
1.4	+3V3A rail	3.3 V ±0.1	U10 TLV75533
1.5	+5V_PAYLOAD rail (PV1)	5.02 V ±0.2, on at power-on	U20 LMR33630ARNXR, L5, divider R42/R43; if dead, Q3 is holding U20’s EN low — check R45
1.6	+3V3_CAN rail	3.3 V ±0.1	U21 XC6206, fed from +5V_PAYLOAD

Raise the limit to 500 mA only once every rail is correct.

If a rail is dead with a correct supply, look at R46 before the buck. It is Q4's gate pull-down. Without it the gate floats, the FET never turns on, and the board is dead with everything wired correctly.

Reverse polarity is protected, but do not test it. Q4 (WST4041 P-FET) blocks a reversed pack and DZ1 clamps the gate, but D1 is still a TVS rather than a series diode. Use the XT60 keying and check twice.

USB powers the logic, not the whole board. D_USB feeds VBUS into the +5 V rail at about 4.7 V, which runs both 3.3 V rails, the MCU and the sensors — enough to configure and flash it. The payload rail stays off (its buck needs about 6.6 V), and through U8's internal diode USB also lifts VBAT, and via Q4 the ESC's battery pin, to about 4 V. Do the first power-up (this Part) on VBAT regardless.

9 Part 8 — T2: MCU alive

A blank STM32H743 already has a bootloader: the ROM DFU loader, which runs when BOOT0 is high through a reset. This board has the button for it (SW1 pulls BOOT0 to 3.3 V; SW2 is reset), so the first flash is over USB and needs no debug probe. Earlier drafts said “bootloader over SWD first, a board with no bootloader will not enumerate” — that was wrong for this MCU. SWD is the *fallback*.

1. Supply on VBAT (Section 8), as the board will be in flight; USB alone would also run the MCU. Data-capable USB-C cable — charge-only leads enumerate nothing.
2. **Hold BOOT (SW1), tap RESET (SW2), release BOOT.** lsusb shows 0483:df11 STM32 BOOTLOADER. If it does not, check Y1 first (the passive crystal — Section 2), then fall back to SWD below.
3. **Flash the ArduPilot bootloader over DFU:** dfu-util -a 0 -s 0x08000000:leave -D Tools/bootloaders/ (built by tools/build_firmware.sh step 3). The board re-enumerates as the ArduPilot bootloader with the board's own ID.
4. Flash arducopter.apj from tools/build_firmware.sh, pinned to Copter-4.7.0, from a ground station or Tools/scripts/uploader.py.
5. Connect a ground station; confirm the board ID and read **every** boot message — every fitted sensor should probe, and nothing unfitted should be claimed.
6. **Copy copter-deadreckon-home.lua to the microSD at APM/scripts/ by hand.** The build script does not package it into the .apj. Without it, FS_EKF_ACTION 0 ships with no GPS-loss response at all.

SWD, if DFU never appears. An ST-Link V2 on J19 (JST-SH 4P: 1 +3V3 as the target-voltage reference, 2 SWDIO, 3 SWCLK, 4 GND) through a 4-pin pigtail — or on the TP1/TP2/TP20/TP21 pads, which carry the same four nets — with st-flash write Tools/bootloaders/NAVCORE_SoOP_b1.bin 0x08000000 or STM32CubeProgrammer. It gets the same bootloader onto the same address; from step 4 on nothing differs.

10 Part 9 — T3: sensors on the bench, props OFF

#	Check	How	Note
3.1	IMU1 (U2) orientation	roll/pitch, watch the horizon	worked out from the pads, never measured
3.2	IMU2 (U3) orientation	same, INS_USE2	only if U3 was fitted

#	Check	How	Note
3.3	Barometer	tracks a 1 m lift	
3.4	Battery voltage	against a meter	BATT_VOLT_MULT 11.000, should be close
3.5	Battery current	known load	ships BATT_AMP_PERVLT 25.0 — a starting point, not a calibration
3.6	RC input	RC_PROTOCOLS 512 (CRSF)	set to 1 for SBUS/FPort
3.7	ESC telemetry	RPM with SERIAL5_PROTOCOL 16	receive-only, and inert on a BL-Heli_S ESC
3.8	Motor order and direction	motor test, props off	J2.3–J2.6 = M1–M4
3.9	Buzzer, LEDs	arming tone, strip lights	
3.10	TFS20-L, if fitted	tape measure at 1 / 5 / 10 m	RNGFND1_TYPE 46
3.11	Compass	calibrate, away from the ESC and the pack	it lives in the GPS module — mount it as far from the 60 A ESC as the frame allows
3.12	Tuner (U13) PLL lock	0x60 on I2C2; tune to 1626 MHz and read the lock-detect bit; a meter on C56 (VTUNE)	VTUNE should sit between the rails, not pinned to 0 V or 3.3 V. Pinned means the loop filter is not doing its job

Everything optional ships disabled in `defaults.parm` and must be turned on here. That is deliberate: a rangefinder configured but absent fails prearm, and a bare board arms out of the box. Do not enable a sensor you have not fitted.

11 Part 10 — T3a: thermal, and the board measures half of it for you

U19 (TMP119) measures U9 for you. It sits 3.9 mm from U9 on I2C1 at 0x48, and `defaults.parm` ships `TEMP_LOG 1`, so every flight writes a TEMP message to the card.

Part	Why it matters	Status
U9 (3.3 V LDO)	linear regulator dropping 1.7 V in SOT-23-5 with no thermal pad, carrying the MCU, flash and microSD	junction 104–157°C against a 150°C limit. It has thermal shutdown — being wrong means the 3.3 V rail switching off in flight, not smoke
U8 (+5 V buck)	16.8 V → 5 V at 0.74 A through a TI LMR33630ARNXR (VQFN-12), swapped in place 2026-09-20, sandwiched between a 60 A ESC and a battery	junction 70–87°C against a 125°C <i>recommended</i> max — 38°C of margin at the worst corner; still an estimate until a thermocouple says so

Both brackets exist because the junction-to-ambient figure is a property of *your* assembled board and the two ends come from different sources (JEDEC and EVM columns of the datasheet).

Why U8's bracket read 70–139°C until 2026-09-20, and what changed. The load it was modelled at, 0.95 A, was a hand-copied budget that still counted the deleted PMW3901/VL53L1X, the DNP CAN transceiver and an eight-sensor ToF ring that the LD06 replaced — 281 mA of parts not on the board — while missing the tuner's 100 mA on U10. `design.LOADS_5V` now derives the rail from the 3.3 V load lists and the fitted sensors: **0.74 A** continuous. The 139°C was retired as a

budget error, and the part was *then* changed — two steps worth keeping apart. The LMR33630 in its 8-pin HSOIC-8 package was measured against the artwork on 2026-09-19 and its courtyard is 0.80 mm too wide for U8’s pocket in every orientation (`docs/KNOWN-ISSUES.md` keeps those numbers); the *same silicon* in the 12-pin RNX VQFN has a 3.9×2.9 mm courtyard, and it was swapped in place on 2026-09-20, taking the worst corner from 117°C to 87°C.

The rule that keeps it there: nothing that is *payload* draws from this board’s buck. The SAWbird+, the recording camera, an FPV AIO and the LD06 run from the **payload 5 V BEC** (stage B2, GBP 4); with the first two alone on the FC rail the worst corner would be 154°C. `check_modules.py` fails any module that is neither in `design.LOADS_5V` nor marked BEC-powered, so a load cannot land on U8 unbudgeted.

Half of that bracket is now measured rather than assumed. Neither package has a case to bolt a heatsink to, so every watt leaves into whatever copper is attached — and `tools/thermal_vias.py` reads that copper off the board for *both* regulators. Measured 2026-09-20, after the swap: U8.1 has **1 via within 1.7 mm holding 7600.3 mm² of GND plane**, U9.2 has 2 vias onto the same plane, and **no legal position exists for another via on either part** (the search runs to 1.7 mm and refuses in-pad vias, which would wick solder). The RNX is a leadframe package with *no* separate exposed pad, so that plane is the whole thermal path — which is what the datasheet’s $R\theta_{JB}$ of 23.3 °C/W is describing. **It still does not license a number** — copper moves the bound monotonically, and only a thermocouple converts it into degrees.

What to do: power it stack-assembled, logging to the card, ten minutes at steady state. Read `TEMP[0]`. If you own a probe, put it on U8 too and on U9’s case once — U19 is 17.2 mm from U8 and cannot speak for it, and one case reading establishes the offset that U19 then carries forward alone.

Stop and ask if `TEMP[0]` exceeds about 85°C or U8’s case exceeds about 100°C.

12 Part 11 — T3b: the RF chain, and the single open question

This is the one place the project is honest about not knowing.

The Doppler solve is written and validated. `tools/soop_solver.py` runs SOLVER OK, recovering a known position from clean geometry at **160.1 m median at 5 Hz**, and reports a singular geometry rather than inventing a number.

What is not known is what the shipping antenna and receiver actually deliver. The shipped `DopplerErrorModel` sigma is **180 m**; the measured divergence boundary was taken at **20 m**. The SITL scenarios print this warning beside their own results:

“The 120 m divergence boundary DOES NOT APPLY here: it was measured at sigma 20 m and this run is at sigma 180 m. Re-measure the boundary at the current sigma before reading a hold/runaway verdict off it.”

Rescaling it by arithmetic would invent a measurement. Only a bench RF measurement moves this: a reference receiver, the real antenna, real geometry. Until then, do not read a position-hold verdict off those scenarios.

13 Part 12 — T3c: IMU temperature calibration

Free, and it attacks the dominant error term: inertial drift is quadratic and dominated by attitude error leaking gravity into the horizontal channel, and gyro and accelerometer bias are strongly temperature-dependent in a stack with almost no airflow. A Pixhawk FMUv6X does this in hardware with IMU heaters; this board (like the MatekH743 it is pin-identical to) has none, so ArduPilot’s software equivalent, `INS_TCAL*`, is the answer. It is deliberately *not* in `defaults.parm`: it is a learning mode, not a flight setting.

1. Run a full six-axis accelerometer calibration first — it sets the `INS_ACC*_CALTEMP` and `INS_GYR*_CALTEMP` the fit is referenced to.

2. Cool the board below its minimum operating temperature: a domestic freezer, in a sealed bag against condensation.
3. Set `INS_TCAL1_ENABLE` 2 (**2 = start learning**, not 1) and `INS_TCAL1_TMAX` to the target; the same for `INS_TCAL2_*` — there are two IMUs.
4. Power it up in a warm room and **do not let it move until the run completes**. Movement invalidates the fit silently.
5. Let it rise through at least 25°C of range, freezer to desk, allowing ten minutes per temperature point. Then verify the fit with the offline plotting tool: the corrected traces should sit near zero across the whole range.

Do this with the **stack assembled**, not the board in free air, because the assembly *is* the thermal environment.

14 Part 13 — T3d: the seven unlabelled pads

Of the **27** pad and test-point references on the board, **20** carry a silkscreen label. A label is printed only where it is legal silkscreen *and* sits nearer its own pad than any other part — `tools/check_silk_owner.py` gates that — because a label that reads as its neighbour’s is worse than a blank pad: it will be trusted. These **7** have no such spot at the board’s own 0.80 mm minimum text height, so they are blank on the board and named on `docs/img/padmap.svg` and the assembly drawings:

P41 P44 P73 PV2 TP1 TP3 TP10

P41 and PZ1 are +5 V; PL2 is +5V_PAYLOAD. TP6 is `USART1_RX`, TP2 is `SWCLK`. Never identify a gold square by counting from its neighbour — use the printed gerber and `docs/img/padmap.svg`. The gate reprints this list on every run, so treat any other number you read as stale.

15 Part 14 — T4/T5: tethered, then flight, strictly in this order

1. **Run the SITL scenarios first:** `bash sitl/run_scenarios.sh`. It found three prearm faults no build check saw.
2. **Battery via the smoke stopper and the EC5-to-XT60 adapter** the first time.
3. **Tethered hover in Stabilize.** Land. Pull the log.
4. Tune `INS_HNTCH_FREQ` and `INS_HNTCH_BW` from the log’s FFT.
5. **GPS Loiter.** This is a working aircraft.
6. **Props on last**, after motor direction is confirmed.

Everything past GPS Loiter is research. GNSS-denied position hold is uncharacterised until T3b is measured.

Operational rule while no flow camera is fitted. EKF source sets 2 and 3 use optical flow for `VELXY`, so today they have **no velocity source**. They are reachable only by the RC switch on `RC9_OPTION` 90: **never select source set 2 or 3 in flight**. Set 1 — GPS, which is also how a SoOP fix will arrive — is the only one with data behind it. And do not “fix” `FLOW_TYPE` 5 to 0 with those sets configured: ArduPilot validates every configured set at arm time, and that would ground the aircraft.

16 Part 15 — Audit: what is still manual, soldered, or unmeasured

This is the audit you asked for. It is split into the three kinds of risk, because each is removed by a different action: *manual work* by buying a jig or an alternative, *hand soldering* by a stencil or an assembly tier, and *unmeasured values* only by a bench session.

16.1 15.1 Manual soldering

The board is machine-assembled, so the soldering is deliberately minimal. What remains:

Item	Joints	Risk and mitigation
Buzzer, J16 or PZ1/PZ2	0–2	low. Must be a passive piezo; fitting an active one costs you every arming and failsafe tone pattern, which you will only notice in the air
LED strip, J15 or PL1/PL2/PL3	0–3	low, and optional. PL1/PL2 are unlabelled — see Section 14
U3, second IMU	14-pin LGA	only if you chose Economic. An LGA-14 has no accessible leads, so it needs hot air or a stencil, not an iron. Standard assembly fits it and removes this entirely

The inefficiency, stated plainly: choosing Economic to save one part moves an LGA-14 onto your bench and does *not* save the feeder fees. It is the single manual soldering task on this board that needs real equipment, and it buys you one optional IMU. Unless you want the redundancy, Standard is the cheaper answer once your time is counted.

There is **no stencil requirement** and no hand-placed passives. That is by design.

16.2 15.2 Manual assembly

Task	Why it is manual, and what would remove it
The stack	four bolts, four spacers, grommets, the ESC and the frame's press nuts. Not automatable and not worth automating
Motor screws	16 screws whose length depends on a tapped depth you must measure. The M3 assortment removes the <i>reorder</i> , not the measurement
Skids	printed, not bought. This is a feature: the 3.5 mm thickness is yours to set, which is why the screw length is not a fixed number
Board connectors	the GPS, RC, I2C, lidar and ESC leads all plug in; no crimping except the spare pigtails
The first flash	the H743's ROM DFU needs only USB and the B00T button (Part 8). SWD — the J19 socket, or the four test pads — is the fallback, not the route
The GPS-loss applet	<code>copter-deadreckon-home.lua</code> must be copied to the microSD by hand; the build does not package it into the <code>.apj</code>
Companion mount	no mount exists yet. Deferred with the companion, so nothing blocks

Two structural constraints worth naming, because neither is removable now. Note that the first reads like a defect and is not — it is the opposite, and worth reading for how it was settled:

- J3's GPS clearance **passes, because the requirement was re-derived rather than the board changed.** The clear run before R21 is **5.30 mm**. The table's old figure of 6.0 mm was a JST-GH *class* nominal, and it assumed a bend allowance that nothing in that corridor needs: a pre-wired GH shell exits its wires at the rear, already parallel to the board, and the only neighbour is a 0402 resistor 0.55 mm tall. Shell half-width 2.6 mm plus 1.0 mm of handling margin is 3.6 mm, so 5.30 mm clears by 1.7 mm. **Relocating R21 was mapped and rejected** — TP20's courtyard and a via fence leave no position in the span that yields 6.0 mm, so there was never a fix available there.

The residual risk is a bulky moulded boot, which is why a spare JST-GH lead is on the order list and T2 fits the plug *before* the headers are soldered.

- **The board is 45.1×47.3 mm, below JLCPCB's 70×70 mm Standard minimum.** Confirmed against their capability page: Standard single-PCB assembly runs 70×70 mm to 460×500 mm, while Economic runs 10×10 mm up. So Standard needs the board panelled with breakaway rails — an extra fabrication step and an extra thing to confirm in their preview, not a blocker. Economic avoids the panel entirely but charges the feeder fees anyway and drops an LGA onto your bench (Section 15.1), which is why Standard plus panelisation is the recommendation.

16.3 15.3 Measurements that have not been confirmed

This is the longest list, and every row is a value some part of the build *assumes*. The gate's own verdict names six of them as NOT READY TO FLY; three more were closed outright on 2026-09-20 by evidence (Section 3.2).

Unconfirmed value	What it decides	Confirmed by
Motor boss tapped depth	M3 $\times$ 14 vs 12 + washer. Too deep a screw clamps nothing while feeling tight	your calipers, before the first of 16 screws. Covered by the assortment
Press-nut tapped depth (30.5 mm joint)	M3 $\times$ 30 vs 28 + washer, on 6.7 mm of engagement	your calipers
Skid thickness, assumed 3.5 mm	sets the motor screw length entirely	you, because you print them. Measure the print
Grommet flange ≤ 5.9 mm	at 6.0 mm it touches a printed label; a 7 mm washer overlaps by 0.52 mm	calipers on the grommets you receive
ESC cable continuity	whether J2 matches the ESC you were sent. Wrong here means a dead board or a smoked one	a multimeter, both ends, before any VBAT
J3 plug against 5.30 mm	whether the received GPS lead seats	fit the plug before the headers are soldered
BATT_AMP_PERVLT	battery failsafe and consumed-mAh. Ships 25.0; it is a property of the ESC's shunt, not of this board	a known load at T3.5
IMU orientations	ROTATION_YAW_270 on both, worked out from the pads and re-checked by <code>check_hwdef.py</code> , never measured. A mirrored axis means it does not fly	roll and pitch it, watch the horizon (T3.1)
U8 junction temperature	70–87°C modelled against a 125°C recommended max, on an honest budget <i>and a swapped part</i> since 2026-09-20; the worst corner has 38°C of margin. If the board measures well past its bracket the fix is a respin	a thermocouple, T3a
U9's board-to-junction offset	makes U19's logged value meaningful rather than a nearby number	one probe reading on the case, then U19 carries it
FLOW_ORIENT_YAW	the sign of the flow correction, if flow is ever fitted	bench calibration

Unconfirmed value	What it decides	Confirmed by
The RF sigma the antenna delivers	everything about GNSS-denied hold. Model says 180 m; the boundary was measured at 20 m	bench RF, T3b. The single open question
Whether this antenna can produce a usable Iridium fix at all	the project's premise	T3b
Flow quality over real grass	simulated flow is perfect flow	a field test
Radxa Zero 3W hole pattern	the companion tray, if the companion is ever fitted	not published — measure the board, then 3D-print a tray
Frame top-plate standoff spacing	nothing to buy: the kit supplies them	the kit
MicroSD withdrawal clearance	how you mount it	your mount
Grommet compression under load	a property of the rubber	the assembled stack

16.4 15.4 The three worst, ranked

1. **The RF sigma (T3b).** It is the only item that can invalidate the project's premise rather than a build step, and no amount of desk work substitutes for it. Everything downstream of it — position-hold claims, the SITL scenarios' hold/runaway results — is explicitly not readable until it is measured.
2. **U8's junction temperature.** 87°C at the worst corner against a 125°C recommended maximum — 38°C of margin, after the 2026-09-20 swap to the LMR33630ARNXR. It is now a number to confirm rather than one to rescue, and the payload BEC is what stops a later addition eating the margin.
3. **The ESC cable and the tapped depths.** Cheap to check, catastrophic to skip. Both are on this list only because they are properties of parts somebody else manufactured, and both are silenced by owning the M3 assortment and a multimeter.

16.5 15.5 Design-level inefficiencies, for a future revision

These do not affect this order. They are recorded because a respin is the only place they can be fixed, and one of them (U8) may force one anyway.

- **USB runs the logic but not the payload rail,** and back-feeds about 4 V onto VBAT and the ESC's battery pin through U8 and Q4.
- **Seven pads arrive unlabelled** (Section 14) because a three-character label at the 0.80 mm minimum height, placed where it reads as its own pad's, has nowhere to go on a 2.84 mm pitch. More silkscreen real estate, or fewer test pads, would remove a bench session.
- **The payload rail cannot carry every payload connector at once** (1.67 A against L5's 1.6 A). A 5 × 5 mm inductor would lift that to 2.1 A, but not on this placement.
- **-nofpv is byte-identical to the default pair.** A variant that ships the same bytes as another is a place for someone to pick the wrong file.
- **The Economic variant saves one part and not the feeder fees,** which makes it a worse deal than it reads as (Section 15.1).
- **In4 rail fragmentation** is an accepted warning on this board.

- **The 1620 MHz SAW filter is not stocked at LCSC**, so the RF front end needs a second supplier and a separate board.

17 Part 16 — What only the outside world can settle

Section 16.3 is the authoritative register of every value this build *assumes*, and it is deliberately not repeated here — an earlier draft restated nine of its rows and the two lists were free to disagree. What follows is the short list no local check and no desk work can close, because each needs a fab, a supplier, a vendor or a maintainer to answer it.

Item	Status
JLPCB's upload preview	outline, panel size, CPL orientation, stackup and drills. No local tool substitutes for looking at their render
1 oz outer copper	every current figure assumes it, and 0.5 oz halves every trace's rating. Confirm what actually ships
LCSC stock	a live external fact, age-gated at 7 days. Re-fetched 2026-09-19 : 56/56 codes resolve, covering 2 boards
The 1620 MHz SAW filter	not stocked at LCSC — a separate board from a separate supplier, and the only part of the RF front end this BOM cannot buy
ArduPilot board ID	7180 is free and below the ceiling; the request, the registry diff and the required <code>Readme.md</code> are ready in <code>firmware/</code> . The local build still uses 9001, which is above the ceiling the registry asks you to respect

18 Part 17 — What is not built yet

The board and its firmware are green. The project's *premise* is not, and a clean gate says nothing about it, so it is stated here rather than left to the repository's own notes.

The Iridium Doppler solver exists and is validated; the chain around it does not.

Piece	Status
The least-squares Doppler solve	built and validated — <code>tools/soop_solver.py</code> . Gauss-Newton over position and clock with an analytic Jacobian; it recovers a known position exactly from clean geometry, and reports <i>singular</i> on degenerate geometry rather than inventing a number
SGP4 and real TLE ephemeris	built and validated (2026-09-20) — <code>tools/soop_solver.py -ephemeris</code> propagates the vendored Iridium NEXT catalogue with SGP4 and converts TEME→ECEF including the Earth's rotation. A known position is recovered from <i>real</i> orbits to ~ 260 m median at 5 Hz over 60 bursts. SGP4 propagation error is not measured by it
Burst detection and frequency estimation	built and validated (2026-09-20) — <code>tools/soop_burst.py</code> , from I/Q: energy detection plus a periodogram carrier estimate that beats the burst's ~ 120 Hz FFT bin, measured at p90 2.4 Hz at $C/N_0 \approx 62$ dB-Hz on synthetic I/Q. The bench re-check on recorded I/Q stays a bench item
The solve ported to C for the H743	built and validated (2026-09-20) — <code>firmware/soop/</code> , a differential test against the Python reference (agreement 1.6×10^{-8} m) and a cross-compile for Cortex-M7
The flight firmware around it	not built — nothing yet samples the MAX2112, runs SGP4 in C, or hands a fix to the EKF on the aircraft
The route into the EKF	built (2026-09-20) — an in-process <code>AP_GPS</code> backend (<code>GPS_TYPE_SOOP</code>), so the fix never leaves the vehicle. Registered and compiled into both firmwares, and gated; that the EKF actually <i>fuses</i> it is not yet proven — <code>tools/soop_sitl_test.py</code> exists but is not passing

The solver also established the receiver's requirement, which was written down nowhere before: **about 5 Hz of burst frequency precision over 40–80 bursts**. An 8.28 ms burst has roughly 120 Hz of raw FFT resolution, so the carrier must be estimated well inside a bin — that is the number T3b is really trying to meet.

19 The one-page order

1. `python3 tools/preflight.py` — must print **READY TO ORDER**
2. `bash tools/make_order_bundle.sh` — regenerate everything
3. Upload `fab/NAVCORE-SoOP-order-bundle.zip`, or the gerbers plus the default BOM/CPL pair
4. 6 layers, 45.1×47.3 mm, 1.6 mm, **1 oz** outer, ENIG, 4 mil, via 0.45/0.20, vias **epoxy filled and capped**, Mark on PCB **Remove Mark**
5. 5 bare boards, 2 assembled. Standard panelised 2×2 unless you want to hand-fit U3
6. Confirm Y1 = C2682774 and Y2 = C5563878; refresh the stock snapshot if it is over 7 days old
7. Order the aircraft parts from `docs/PARTS.csv`, **including the M3 assortment**, the SAWbird+ with its coax pigtail and power lead, and the bias-tee dongle. Do not buy a bare PMW3901 or VL53L1X chip
8. While waiting: prepare the board ID request, print the padmap, decide the USB edge
9. On arrival: inspect under magnification, buzz the ESC cable, measure the tapped depths
10. T1 current-limited rails → T2 bootloader over USB DFU (BOOT + RESET) → T3 sensors, props off
11. T3a thermal (U8 needs a probe) → T3c IMU temp cal → T3d seven unlabelled pads
12. T3b before any position-hold claim: measure the real sigma
13. SITL scenarios → tethered hover → GPS Loiter. **Props on last**

Generated from the NAVCORE-SoOP working tree, 19 September 2026.

Gate: READY TO ORDER, every gated prerequisite proven — the count is printed by the gate, not typed here.

Hardware: NOT READY TO FLY, 6 items requiring hardware.